

March 21 - Janvier - 2025

② Pg 174; Ex. 318

$$a) P(x) = 4x^4 - 15x^3 + 20x^2 - 10x + 14 \quad a = 1$$

$$P(x) = P(1) + \frac{P'(1)(x-1)}{1!} + \frac{P''(1)(x-1)^2}{2!} + \frac{P'''(1)(x-1)^3}{3!} + \frac{P^{(4)}(1)(x-1)^4}{4!}$$

$$P(1) = 4(1)^4 - 15(1)^3 + 20(1)^2 - 10(1) + 14$$

$$P(1) = 13$$

$$P'(x) = 16x^3 - 45x^2 + 40x - 10 \quad P'(1) = 16 - 45 + 40 - 10 = -5$$

$$P'(1) = 16(1)^3 - 45(1)^2 + 40(1) - 10 \quad P'(1) = -5$$

$$P(1) = 1$$

$$P''(x) = 48x^2 - 90x + 40$$

$$P''(1) = 48(1)^2 - 90(1) + 40$$

$$P''(1) = -2$$

$$P'''(x) = 96x - 90$$

$$P'''(1) = 96(1) - 90$$

$$P'''(1) = 6$$

$$P(x) = 13 + \frac{-5}{1!}(x-1) + \frac{-2}{2!}(x-1)^2 + \frac{6}{3!}(x-1)^3 + \frac{96}{4!}(x-1)^4$$

$$P(x) = 13 - 5(x-1) - (x-1)^2 + (x-1)^3 + 4(x-1)^4$$

$$b) P(x) = x^3 + 4x^2 - 5x + 7 \quad a = -2$$

$$P(x) = P(-2) + \frac{P'(-2)}{1!} (x+2) + \frac{P''(-2)}{2!} (x+2)^2 + \frac{P'''(-2)}{3!} (x+2)^3$$

$$P(-2) = (-2)^3 + 4(-2)^2 - 5(-2) + 7$$

$$P(-2) = 95$$

$$P'(x) = 3x^2 + 8x - 5$$

$$P''(x) = 6x + 8$$

$$P''' = 6$$

$$P'(-2) = 3(-2)^2 + 8(-2) - 5 \quad P''(-2) = 6(-2) + 8 \quad P'''(-2) = 6$$

$$P'(-2) = -9$$

$$P''(-2) = -4$$

$$P(x) = 95 + \frac{-9}{1!} (x+2) + \frac{-4}{2!} (x+2)^2 + \frac{6}{3!} (x+2)^3$$

$$P(x) = 95 - 9(x+2) - 2(x+2)^2 + (x+2)^3$$

$$3) P(x) = 7x^5 - 8x^4 + x^3 - 2x^2 + 5x - 7 \quad a = 0$$

$$P(x) = P(0) + \frac{P'(0)}{1!} (x) + \frac{P''(0)}{2!} (x)^2 + \frac{P'''(0)}{3!} (x)^3 + \frac{P^{(4)}(0)}{4!} (x)^4 + \frac{P^{(5)}(0)}{5!} (x)^5$$

$$P(0) = 7(0)^5 - 8(0)^4 + (0)^3 - 2(0)^2 + 5(0) - 7$$

$$P(0) = -7$$

$$P'(x) = 35x^4 - 32x^3 + 3x^2 - 4x + 5 \quad P^{(4)}(x) = 840x - 192$$

$$P'(0) = 5$$

$$P^{(4)}(0) = -192$$

$$P''(x) = 140x^3 - 96x^2 + 6x - 4$$

$$P^{(5)}(x) = 840$$

$$P''(0) = -4$$

$$P^{(5)}(0) = 840$$

$$P^{(3)}(x) = 420x^2 - 192x + 6$$

$$P^{(3)}(0) = 6$$

$$P(x) = -1 + \frac{5}{1!}x + \frac{-4}{2!}x^2 + \frac{6}{3!}x^3 + \frac{-19}{4!}x^4 + \frac{84}{5!}x^5$$

$$P(x) = -1 + 5x - 2x^2 + x^3 - 8x^4 + 7x^5$$

④ Pg. 177 Ex. 388

$$f(x) = x^2 \cdot e^x$$

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$$

$$f'(x) = (x^2)' e^x + x^2 (e^x)'$$

$$f'(x) = 2x \cdot e^x + x^2 \cdot e^x$$

$$f'(x) = (x^2 + 2x) e^x$$

$$\begin{aligned} f''(x) &= (x^2 + 2x)' e^x + (x^2 + 2x) (e^x)' \\ &= (2x + 2) e^x + (x^2 + 2x) e^x \end{aligned}$$

$$f''(x) = (x^2 + 4x + 2) e^x$$

$$f''(x) = (2x + 4x + 2) e^x + (x^2 + 4x + 2) e^x$$

$$f''(x) = (x^2 + 6x + 6) e^x$$

$$f''(x) = (x^2 + 6x + 6) e^x$$

$$f^{(4)}(x) = (x^2 + 6x + 6)' e^x + (x^2 + 6x + 6) (e^x)'$$

$$f^{(4)}(x) = (2x + 6) e^x + (x^2 + 6x + 6) e^x$$

$$f^{(4)}(x) = (x^2 + 8x + 12) e^x$$

At $x=0$, $f^{(4)}(0) = (0^2 + 8 \cdot 0 + 12) \cdot e^0 = 12$

$$\frac{f^{(4)}(0)}{4!} = \frac{12}{4!} = \frac{1}{2} \quad \text{Ans. 5}$$

Mercrèdi, 22 - Janvier - 2025

⑤ Pg. 176 ex. 398

$$P(x) = 3x^7 + 2x^6 + 4x^3 - 3x^2 + 5x + 1 \quad a = 1$$

$$\frac{P''(1)}{3!} + \frac{P^{(7)}(1)}{7!} = ?$$

$$P(x) = P(1) + \frac{P'(1)}{1!}(x-1) + \frac{P''(1)}{2!}(x-1)^2 + \frac{P'''(1)}{3!}(x-1)^3 + \frac{P^{(4)}(1)}{4!}(x-1)^4 +$$

$$\frac{P^{(5)}(1)}{5!}(x-1)^5 + \frac{P^{(6)}(1)}{6!}(x-1)^6 + \frac{P^{(7)}(1)}{7!}(x-1)^7$$

$$P'(x) = 21x^5 + 12x^5 + 12x^2 - 6x + 5$$

$$P''(x) = 106x^4 + 60x^4 + 24x - 6$$

$$P'''(x) = 424x^3 + 240x^3 + 24 \quad P'''(1) = 424(1)^3 + 240(1)^3 + 24 = 884$$

$$P^{(4)}(x) = 1272x^2 + 720x^2$$

$$P^{(5)}(x) = 2544x + 720$$

$$P^{(6)}(x) = 2544$$

$$P^{(7)}(x) = 2544$$

$$P^{(7)}(1) = 2544$$

$$\frac{P'''(1)}{3!} + \frac{P^{(7)}(1)}{7!} = \frac{884}{3!} + \frac{2544}{7!} = \frac{884}{6} + \frac{2544}{5040} = \underline{\underline{159}}$$

⑥ Déterminer le développement au voisin de 1 de la fonction

$$f(x) = (1+x)^k \quad (k \in \mathbb{R}^*)$$

$$f(0) = (1+0)^k = 1$$

$$f'(x) = k(1+x)^{k-1} \quad f'(0) = k(1+0)^{k-1} = k$$

$$f''(x) = k(k-1)(1+x)^{k-2} \quad f''(0) = k(k-1)(1+0)^{k-2} = k(k-1)$$

$$f'''(x) = k(k-1)(k-2)(1+x)^{k-3} \quad f'''(0) = k(k-1)(k-2)(1+0)^{k-3} = k(k-1)(k-2)$$

$$f^{(4)}(x) = k(k-1)(k-2)(k-3)(1+x)^{k-4} \quad f^{(4)}(0) = k(k-1)(k-2)(k-3)$$

$$f^{(5)}(0) = k(k-1)(k-2)(k-3)(k-4)$$

$$\text{D'où: } (1+x)^k = 1 + \frac{k}{1!} x + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \frac{k(k-1)(k-2)(k-3)}{4!} x^4 + \dots$$

① Déterminer suivant le développement de $(1+x)^k$, la valeur approchée de $(1,0003)^5$ à 10^{-3} près.

$$(1,0003)^5 = (1 + 0,0003)^5 = (1 + 3 \cdot 10^{-4})^5 \quad 5 = k$$

$$1 + \frac{5(3 \cdot 10^{-4})}{1!} + \frac{5(5-1)(3 \cdot 10^{-4})^2}{2!} + \frac{5(4)(9 \cdot 10^{-8})}{3!} = 1 + 15 \cdot 10^{-4} + 10 \cdot 9 \cdot 10^{-8} = 1 + 15 \cdot 10^{-4} + 90 \cdot 10^{-8}$$

$$1 + 15 \cdot 10^{-4} + 90 \cdot 10^{-8} = 1,0015 \quad = 10 \cdot 9 \cdot 10^{-8} = 90 \cdot 10^{-8}$$

$$= 1,001$$

② Pg 177 Ex. 339.

$$f(x) = \sqrt[3]{1,1} + 0,139$$

$$= (1,1)^{1/3} + 0,139$$

$$= (1 + 0,1)^{1/3} + 0,139$$

$$= 1 + \frac{1/3 \cdot 0,1}{1!} + \frac{1/3(1/3-1)(0,1)^2}{2!} + \frac{1/3(1/3-1)(1/3-2)(0,1)^3}{3!} + 0,139$$

$$= 1 + 0,033 + \frac{1/3 \cdot (-2/3)(0,1)^2}{2} + \frac{1/3 \cdot (-2/3) \cdot (-5/3)(0,1)^3}{6} + 0,139$$

$$= 1,033 - 1,1 \cdot 10^{-3} + 6,17 \cdot 10^{-5} + 0,139$$

$$= 1,1639 \approx 1,164 \quad \text{AM:5}$$

⑤ Pg. 177. Ex. 342

$$f(x) = \ln(x + \sqrt{1+x^2}) \quad f(x) = ax + bx^2 + cx^3$$

$$a - b + c = ?$$

$$\textcircled{1} f'(x) = \frac{(x + \sqrt{1+x^2})'}{x + \sqrt{1+x^2}} = \frac{1 + \frac{(1+x^2)'}{2\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}(x + \sqrt{1+x^2})}$$

$$f'(x) = \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}} \cdot \frac{1}{x + \sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}} = \frac{1}{(1+x^2)^{1/2}} = (1+x^2)^{-1/2}$$

$$f'(0) = \frac{1}{\sqrt{1+0}} = 1 \quad \underline{\underline{f'(0) = 1}}$$

$$\textcircled{2} f''(x) = [(1+x^2)^{-1/2}]' = -\frac{1}{2} (1+x^2)^{-1/2-1} (1+x^2)'$$

$$= -\frac{1}{2} (1+x^2)^{-3/2} \cdot 2x = -x (1+x^2)^{-3/2}$$

$$\underline{\underline{f''(x) = -x (1+x^2)^{-3/2}}} \quad f''(0) = -0 \cdot (1+0)^{-3/2} \quad \underline{\underline{f''(0) = 0}}$$

$$\textcircled{3} f'''(x) = [-x (1+x^2)^{-3/2}]' = (-x)' (1+x^2)^{-3/2} + (-x) [(1+x^2)^{-3/2}]'$$

$$= -1 (1+x^2)^{-3/2} - x \left[-\frac{3}{2} (1+x^2)^{-3/2-1} \cdot (1+x^2)' \right]$$

$$= - (1+x^2)^{-3/2} - x \left[\frac{3 (1+x^2)^{-5/2} \cdot 2x}{2} \right]$$

$$\underline{\underline{f'''(x) = - (1+x^2)^{-3/2} + 3x^2 (1+x^2)^{-5/2}}}$$

$$f'''(0) = - (1+0)^{-3/2} + 3 \cdot 0 \cdot (1+0)^{-5/2}$$

$$\underline{\underline{f'''(0) = -1}}$$

$$a = \frac{f'(0)}{1!} = \frac{1}{1} = 1 ; b = \frac{f''(0)}{2!} = \frac{0}{2} = 0 ; c = \frac{f'''(0)}{3!} = \frac{1}{6}$$

$$\textcircled{10} \text{ D'où : } a + b + c = 1 + 0 + \frac{1}{6} = \underline{\underline{\frac{5}{6}}}$$

⑩ P178 Ex. 351.

$$f(x) = e^{2x}$$

Le terme en x^8 du développement selon M-L est de la forme :

$$\frac{f^{(8)}(0)}{8!} x^8$$

$$f'(x) = (e^{2x})' = (2x)' e^{2x} = 2e^{2x} \quad | \quad f^{(8)}(0) = 256 e^{2 \cdot 0} = 256 \cdot 1 = 256$$

$$f''(x) = (2e^{2x})' = 4e^{2x}$$

$$f'''(x) = (4e^{2x})' = 8e^{2x}$$

$$f^{(4)}(x) = (8e^{2x})' = 16e^{2x}$$

$$f^{(5)}(x) = (16e^{2x})' = 32e^{2x}$$

$$f^{(6)}(x) = (32e^{2x})' = 64e^{2x}$$

$$f^{(7)}(x) = (64e^{2x})' = 128e^{2x}$$

$$f^{(8)}(x) = (128e^{2x})' = 256e^{2x}$$

$$\text{Donc : } \frac{f^{(8)}(0)}{8!} x^8 = \frac{256}{40320} x^8$$

$$= \underline{\underline{\frac{2}{315} x^8}}$$

⑩ Pg. 176 Ex 333

① $F(x) = (2,0003)^5$ à 10^{-3} près

$F(x) = (2+0,0003)^5$ d'où $F(x) = (2+3 \cdot 10^{-4})^5$ avec $k=5$ et $x = 3 \cdot 10^{-4}$.

$F(x) = (2+3 \cdot 10^{-4})^5$ / $F(x) = (2+x)^k$

$F(x) = (2+x)^k$

$F'(x) = k(2+x)^{k-1}$

$F''(x) = k(k-1)(2+x)^{k-2}$

$F'''(x) = k(k-1)(k-2)(2+x)^{k-3}$

$F^{(4)}(x) = k(k-1)(k-2)(k-3)(2+x)^{k-4}$

$F^{(5)}(x) = k(k-1)(k-2)(k-3)(k-4)(2+x)^{k-5}$

$F(x) = (2+x)^k = 2^k$

$F'(x) = k(2+x)^{k-1} = k \cdot 2^{k-1}$

$F''(x) = k(k-1)(2+x)^{k-2} = k(k-1) \cdot 2^{k-2}$

$F'''(x) = k(k-1)(k-2)(2+x)^{k-3} = k(k-1)(k-2) \cdot 2^{k-3}$

$F^{(4)}(x) = k(k-1)(k-2)(k-3)(2+x)^{k-4} = k(k-1)(k-2)(k-3) \cdot 2^{k-4}$

$F^{(5)}(x) = k(k-1)(k-2)(k-3)(k-4)(2+x)^{k-5} = k(k-1)(k-2)(k-3)(k-4) \cdot 2^{k-5}$

$F(x) = F(0) + \frac{F'(0)}{1!}x + \frac{F''(0)}{2!}x^2 + \frac{F'''(0)}{3!}x^3 + \frac{F^{(4)}(0)}{4!}x^4 + \frac{F^{(5)}(0)}{5!}x^5$

② Valeur approximative de $(2,0003)^5$ à 10^{-3} près.

$F(x) = (2+3 \cdot 10^{-4})^5$ $k=5$ et $x = 3 \cdot 10^{-4}$

$F(x) = F(0) + \frac{F'(0)}{1!}x + \frac{F''(0)}{2!}x^2 + \frac{F'''(0)}{3!}x^3 + \dots$

$F(0) = 32$

$F'(0) = 80$

$F''(0) = 160$

$F'''(0) = 240$

$F(x) = 32 + \frac{80}{1!}(3 \cdot 10^{-4}) + \frac{160}{2!}(3 \cdot 10^{-4})^2 + \frac{240}{3!}(3 \cdot 10^{-4})^3 + \dots$

$F(x) = 32 + 24 \cdot 10^{-3} + 7,2 \cdot 10^{-8} + 1,08 \cdot 10^{-9} + \dots$

$F(x) = 32,024$

✓ 9/5